PARTH PATKAR

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SUMMARY

AI/ML Engineer with hands-on experience in **Python, SQL, Machine Learning, Generative AI, and FastAPI**. Skilled in developing **LLM-powered applications, RAG pipelines, REST APIs**, and predictive models using **LangChain, LangGraph,** and Scikit-learn. Experienced in building AI-driven automation solutions and deploying intelligent applications.

TECHNICAL SKILLS

- **Programming & Data:** Python (Pandas, NumPy, Scikit-learn), SQL, MySQL, Excel, Power BI, Tableau
- **AI & Machine Learning:** Regression, Classification, Deep Learning, GenAI, Agentic AI, LangChain, LangGraph
- **Deployment & Cloud:** FastAPI, REST API , Pydantic.

PROFESSIONAL EXPERIENCE

HCLTech | September 2026 – Present | Madurai, India

- Undergoing technical upskilling in Python, SQL, REST APIs, FastAPI, and Generative AI, with a focus on Microsoft AI Foundry, LangChain, and LangGraph for AI-powered application development and enterprise solutions.

PROJECTS

AI-Powered ServiceNow Incident Automation (Prototype) | Personal Project
Python, FastAPI, SQLAlchemy, SQLite, REST APIs, Ollama (LLM), RAG, ServiceNow Table API, Streamlit, Pytest

- Architected an end-to-end incident automation pipeline in **Python/FastAPI** with full **CRUD REST APIs,** a rule-based + LLM classification engine (**Ollama**), and a **RAG** retrieval layer combining knowledge-base articles and historical ticket data to generate structured AI recommendations (category, priority, root cause, confidence score) gated behind human-in-the-loop approval
- Integrated the system with the ServiceNow REST Table API for live incident retrieval and work-note updates, built a **Streamlit** dashboard for automation-opportunity analysis across **1,000+** synthetic tickets, and validated reliability with a **Pytest** suite covering CRUD, classification, and AI confidence-gating logic (**12/12 passing**)

PARTH.AI — AI-Powered Portfolio Assistant | React.js, FastAPI, Python, Groq API, RAG

- Architected an LLM-powered portfolio assistant using **React.js, FastAPI, Groq API, and RAG**, integrating knowledge retrieval across multiple Markdown documents with real-time SSE streaming.
- Optimized the **RAG architecture** by replacing **LangChain, ChromaDB**, and embedding-model dependencies with **lightweight Python-based retrieval**, **reducing infrastructure complexity** and enabling deployment on resource-constrained Render hosting.

Rainfall Prediction using Machine Learning | Python, Scikit-learn, XGBoost, Imbalanced-learn

- Developed and optimized end-to-end ML pipelines for rainfall classification on a 366-sample dataset, implementing **outlier handling, feature scaling, SMOTE, and PCA** to improve prediction performance.
- Benchmarked Logistic Regression, Random Forest, and XGBoost using GridSearchCV and Repeated Stratified K-Fold, improving test accuracy from **74.5% to 77%** and cross-validation accuracy from **82% to 87%**, while analyzing precision-recall trade-offs.

EDUCATION

Malaviya National Institute Of Technology , Jaipur.
Metallurgical and Material Science | CGPA : **8.31** (2022-2026)

Gogate Joglekar College .
Science 12th Marks : **85.50%** (2020-2022)

KMSP Mandal English Medium School Kudal.
10th Marks: **94.40%** (2019-2020)

EXTRA CURRICULAR

- Led **20+ member** creative design team for MNIT's annual fest, ensuring cohesive branding and smooth execution.
- **Managed logistics** for **30+ speakers & 200+ attendees** at an international conference, **coordinating travel, accommodation,** and scheduling for seamless event operation
- **Managed logistics and décor** for Sphinx'23 and Blitzchalag'23 as an Executive, leading a **15+ member team** to ensure seamless execution and visually engaging event setups.